

1. Create a comparison table listing the main differences between DDR3, DDR4, and DDR5 RAM, including speed, voltage, and typical use cases.

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DDR3 vs DDR4 vs DDR5 – Key Differences			
Feature	DDR3	DDR4	DDR5
Release Year	2007	2014	2020
Speed (Typical Range)	800 – 2133 MT/s	2133 – 3200 MT/s	4800 – 9600+ MT/s
Data Transfer Rate (MT/s)	Up to 2133 MT/s	Up to 3200 MT/s	4800 MT/s and higher
Voltage	1.5V (1.35V low voltage)	1.2V	1.1V
Max Capacity per DIMM (Typical)	8GB	16GB (some up to 32GB)	32GB – 64GB
Pins	240-pin	288-pin	288-pin
Architecture / Bank Groups	8 Banks	16 Banks	32 Banks
ECC Support	Yes (on some modules)	Yes (on some modules)	Yes (on-die ECC for better reliability)
Power Efficiency	Lower	Better	Best
Typical Use Cases	Older desktops, laptops, budget systems	Mainstream desktops, laptops, gaming PCs (2014–2020 era)	High-performance PCs, gaming, content creation, AI, future systems
Compatibility	Works only with DDR3 motherboards	Works only with DDR4 motherboards	Works only with DDR5 motherboards

2. Open your laptop or desktop (with power disconnected), locate the RAM slots, and take a clear photo of the installed RAM module. Reinstall the RAM module and verify that your system boots up successfully.

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can complete the task as follows:

1. **Shut down the computer** and unplug the power cable.
2. Press the power button for a few seconds to discharge any remaining power.
3. Open the case (desktop) or service panel (laptop).
4. Locate the **RAM slots (DIMM/SODIMM slots)**.
5. Take a clear photo of the installed RAM module.
6. Release the retaining clips and carefully remove the RAM.
7. Reinstall the RAM by aligning the notch and pressing firmly until the clips lock into place.
8. Reconnect power and start the computer.
9. Verify that the system boots normally and that the correct amount of RAM is detected.

3. List three key differences between HDD, SSD, and NVMe storage types, and for each, name one app or use case (from apps you use daily) that would benefit most from that storage type.

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વિશેષતા	HDD	SSD	NVMe SSD
સ્પીડ	ધીમી	ઝડપી	સૌથી ઝડપી
ટેકનોલોજી	ફરતી ડિસ્ક (Mechanical)	Flash Memory	Flash Memory + PCIe Interface
કિંમત	સસ્તી	મધ્યમ	સૌથી મોંઘી

Storage Type	શ્રેષ્ઠ ઉપયોગ	App/Use Case
HDD	મોટા પ્રમાણમાં ફાઇલો અને વિડિઓ સ્ટોર કરવા	Movies, Photos Backup
SSD	ઝડપી બૂટ અને સામાન્ય કામ	Chrome, Spotify, MS Office
NVMe SSD	હાઈ-સ્પીડ ગેમિંગ અને પ્રોફેશનલ કામ	PUBG PC, FIFA, Adobe Premiere Pro

4. Remove the storage drive (HDD or SSD) from your system, take a photo of the connector type (SATA, M.2, or NVMe), and reinstall the drive. Confirm that your device recognizes the storage after reassembly.
- Constraint:** Only perform this task if you have access to a device you are allowed to open. If not, find and share a YouTube video link showing the process for your laptop/PC model.

Topic Repository URL

=><https://youtu.be/gp4uizM96EE?si=Mqe640J6WZDkWaLY>